

Vidath Dassanayake

COMPUTER SCIENCE UNDERGRADUATE | SOFTWARE ENGINEERING

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PROFESSIONAL SUMMARY

Computer Science undergraduate with Python and TypeScript experience, who built and deployed a full-stack SaaS platform (Next.js, Express, PostgreSQL) with a 6-person Agile team. Seeking a 6-month software engineering internship.

EDUCATION

Informatics Institute of Technology (IIT) – BSc (Hons) Computer Science · Year 3 of 4 · Expected graduation: 2024 – Present 2028

Modules – Object-Oriented Programming (Java/Python, Data Structures & Algorithms), Databases, Client-Server Architecture, Web Development

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C# (Unity), HTML, CSS, OOP
Frameworks & Libraries: Next.js, React, Node.js, Express.js, FastAPI, Prisma ORM, PyTorch
Databases: PostgreSQL, Redis, SQL
Cloud & DevOps: Docker, Kubernetes, GKE, GCP, Terraform, Helm, GitHub Actions, Kong
Other: Linux, Bash, Git, Postman, Unity 6, Blender

PROJECTS

OreanStudio (Orean 360 Platform)

2025 – 2026

Technologies: Next.js, React, Node.js 20, Express.js, TypeScript, Zod, PostgreSQL, Prisma ORM, BullMQ, Redis, FFmpeg, Kubernetes, GKE, Terraform, Helm, Kong API Gateway, GitHub Actions

- Built a Next.js 16 frontend and a Node.js 20 / Express 5 REST API in TypeScript (Prisma ORM, JWT auth, Zod validation) on a 14-model PostgreSQL schema.
- Implemented six OAuth integrations (Google, YouTube, Meta, LinkedIn, TikTok, WhatsApp), with BullMQ workers for scheduled publishing (retries with exponential backoff, publish audit logs) and FFmpeg video transcoding into five platform presets.
- Built the post-creation, Kanban board, due-date update and calendar date-range endpoints, plus social-token disconnection.
- Automated Prisma migrations and Helm deployments to GKE through GitHub Actions CI/CD (with atomic rollback), and implemented KEDA-driven autoscaling for BullMQ workers.
- Tested the platform manually, including cross-browser (Chrome, Firefox, Edge) and responsive testing that found dashboard layout defects on small screens.
- Defined 10 prioritised functional requirements from a 14-person industry survey and 2 stakeholder interviews.
- Delivered the platform with a 6-member team using Agile/Scrum practices, and validated it through feedback from end users, a domain expert, industry experts and the IIT project showcase.

GEO Engine (Zero-Cost LLM Pipeline)

2025 – 2026

Technologies: Python, PyTorch, Unsloth, Qwen2.5, LoRA, vLLM, Firecrawl, Playwright, Docker Compose, WSL2, Modal

- Built a full-stack GEO (Generative Engine Optimisation) application that optimises content for LLMs like ChatGPT and Gemini.
- Cleaned the scraped data (minimum-length filter, deduplication, spam and all-caps removal, PII stripping) down to 374 records and formatted them as ChatML, split 299/37/38 into train, validation and test sets.
- Fine-tuned Qwen2.5-7B-Instruct with LoRA (Unsloth, 4-bit) on a free Google Colab T4 GPU; training loss fell from 1.84 to 0.96 over 3 epochs at \$0 compute cost.
- Served the merged model with vLLM on Modal (A10G GPU, scale-to-zero after 120 seconds idle) behind an OpenAI-compatible API used by the Orean backend, within Modal's free credit tier.
- Built a rate-limited data-collection Python pipeline with self-hosted Firecrawl in Docker; gathered 377 programming posts from 15 subreddits (score ≥ 100 , upvote ratio $\geq 85\%$) through the Arctic Shift archive.

Modly Blender Extension

2025

Technologies: Python, Blender Python API (bpy), FastAPI

- Developed a Blender 5.2 add-on integrating Modly's local AI 3D generation via a custom Node Graph, with a dynamic node tree generated from JSON manifests and an isolated FastAPI backend to prevent GPU conflicts.

CERTIFICATIONS

IBM SkillsBuild: [Cloud Computing Fundamentals](#)

Cisco: [Python Essentials 2](#)

Boot.dev: [Docker](#), [Git](#), [Linux](#), [SQL](#), [Python](#), [OOP in Python](#)

DataCamp: Java, Java OOP

LANGUAGES

English: Fluent
Sinhala: Native